

COURSE MATERIAL

SUBJECT	ENGINEERING GRAPHICS (ME23AES102)
UNIT	1
COURSE	B.TECH
SEMESTER	11
DEPARTMENT	MECHANICAL ENGINEERING
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1. COURSE OBJECTIVES

The objectives of this course is to introduce and make the students

1. To enable with various concepts like dimensioning, conventions and standards related to Engineering Drawing
2. To impart knowledge on the Projection of Points, Lines and Plane surfaces
3. To improve the visualization skills for better understanding of projection of solids
4. To develop the imaginative skills required to understand Section of solids and Developments of surfaces.
5. To understand the viewing perception of a solid object in Isometric and orthographic projections.

2. PREREQUISITES

Students should have knowledge on
Basic Mathematics

3. SYLLABUS**UNIT I**

Introduction to Engineering Drawing: Lines, Lettering and Dimensioning, Geometrical Constructions and Constructing Regular Polygons by General Methods.

Curves: Construction of Ellipse, Parabola and Hyperbola by General Method, Cycloids, Involute, Normal and Tangent to Curves.

Scales: Plain Scales, Diagonal Scales and Vernier Scales.

4. COURSE OUTCOMES

After completing the course, the student will be able to

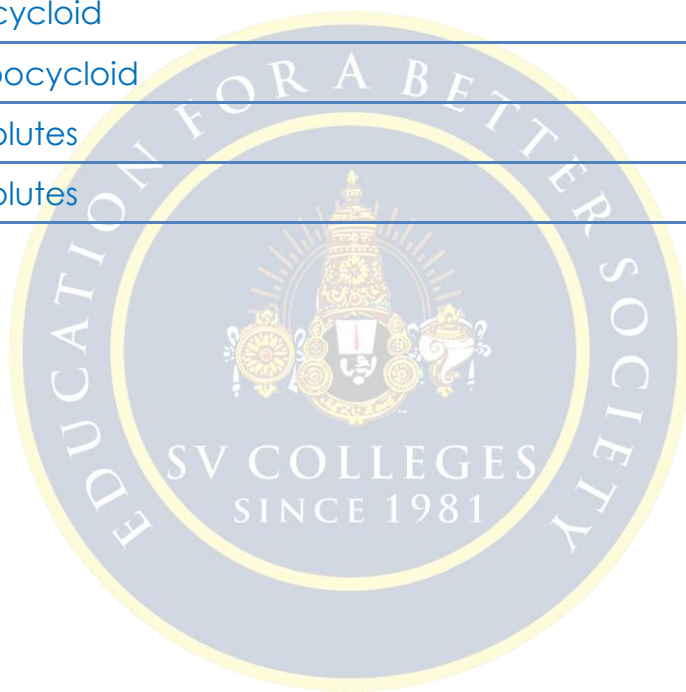
1. Draw various engineering curves, scales. (L3)
2. Draw and Interpret orthographic projections of points, lines, planes. (L3)
3. Draw the projection of solids in various positions. (L3)
4. Draw and Explore the sections of solids and development of surfaces. (L3)
5. Draw an isometric and orthographic views of simple solids. (L3)

5. Co-PO / PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	PO11	PO12	PSO1	PSO2
CO1	2	2								2		2	2	2
CO2	2	2								2		2	2	2
CO3	2	2								2		2	2	2
CO4	2	2								2		2	2	2
CO5	2	2			1					2		2	2	2

6. LESSON PLAN

LECTURE	WEEK	TOPICS TO BE COVERED	REFERENCES
1	1	Principles of Engineering Drawing	T1
2		Conventions in drawing lettering BIS conventions	T1, R1
3		Conic sections	T1, R1
4		Conic sections	T1, R2
5	2	Cycloid	T1, R1
6		Epicycloid	T2, R1
7		Hypocycloid	T2, R1
8	3	Involutes	T2, R1
9		Involutes	T2, R2



7. LECTURE NOTES

1.1 INTRODUCTION

Engineering drawing is a two dimensional representation of three dimensional objects. In general, it provides necessary information about the shape, size, surface quality, material, manufacturing process, etc., of the object. It is the graphic language from which a trained person can visualize objects.

1.1.1 DRAWING INSTRUMENTS AND AIDS:

The Instruments and other aids used in drafting work are listed below:

1. Drawing board
2. Set squares
3. French curves
4. Templates
5. Mini drafter
6. Instrument box
7. Protractor
8. Set of scales
9. Drawing sheets
10. Pencils

1.1.2 DRAWING BOARD:

Until recently drawing boards used are made of well-seasoned softwood of about 25 mm thick with a working edge for T-square. Nowadays mini-drafters are used instead of T-squares which can be fixed on any board. The standard size of board depends on the size of drawing sheet size required.

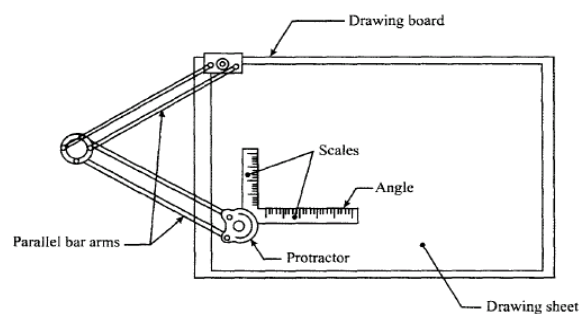


Fig.1.1 Mini Draughter

1.1.3 MINI-DRAFTER:

Mini-drafter consists of an angle formed by two arms with scales marked

and rigidly hinged to each other. It combines the functions of T-square, set-squares, scales, and protractor. It is used for drawing horizontal, vertical, inclined lines, parallel and perpendicular lines and for measuring lines and angles.

1.1.4 INSTRUMENT BOX

Instrument box contains 1. Compasses, 2. Dividers and 3. Inking pens. What is important is the position of the pencil lead with respect to the tip of the compass. It should be at least 1 mm above as shown in the fig. because the tip goes into the board for grip by 1 mm.

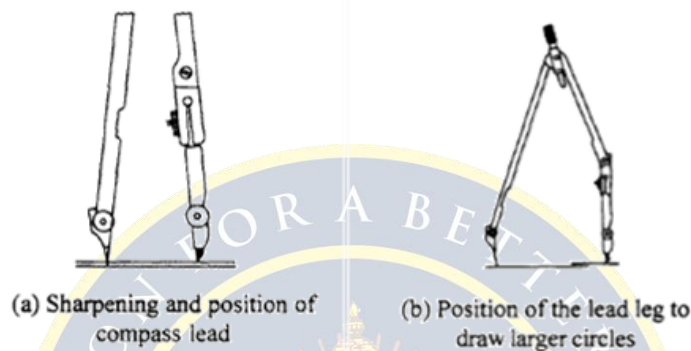


Fig.1.2 Compass and its position

1.1.5 PENCILS:

Pencils with leads of different degrees of hardness or grades are available in the market. The hardness or softness of the lead is indicated by 3H, 2H, H, HB, B, 2B, 3B, etc. The grade HB denotes medium hardness of lead used for general purpose. The hardness increases as the value of the numeral before the letter H increases. The lead becomes softer, as the value of the numeral before B increases.

- HB Soft grade for Border lines, lettering, and free sketching
- H Medium grade for Visible outlines, visible edges, and boundary lines
- 2H Hard grade for construction lines, Dimension lines, Leader lines, Extension lines, Centre lines, Hatching lines and Hidden lines.

1.1.6 Drawing Sheet

The standard drawing sheet sizes are arrived at on the basic Principle of $x:y = 1:2^{(1/2)}$ and $xy = 1$ where x and y are the sides of the sheet. For example, AO, having a surface area of 1 Sq.m; $x = 841$ mm and $y = 1189$ mm. The successive sizes are obtained by either by halving along the length or doubling the width, the area being in the ratio 1: 2. Designation of sizes is given in the fig.

For class work use of A2 size drawing sheet is preferred.

Table.1.1 Designation of different sheets Fig.1.3 A2 size drawing sheet

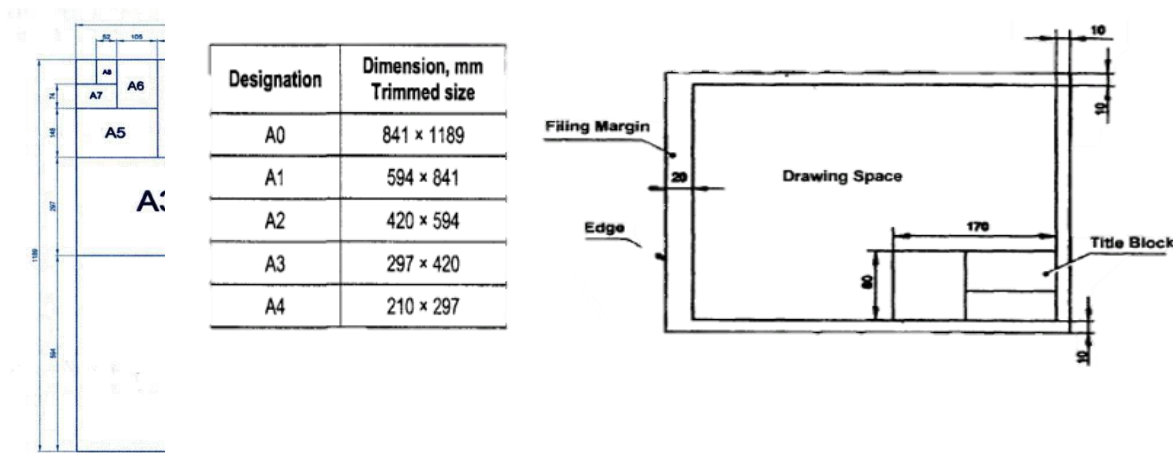


Fig.1.4 A2 size drawing sheet

1.1.7 TITLE BLOCK:

The title block should lie within the drawing space at the bottom right hand corner of the sheet. The title block can have a maximum length of 170 mm and width of 65mm providing the following information.

- Title of the drawing.
- Drawing number.
- Scale.
- Symbol denoting the method of projection.
- Name of the firm, and
- Initials of staff, who have designed, checked, and approved.

1.1.8 LINES:

Just as in English textbook the correct words are used for making correct sentences; in Engineering Graphics, the details of various objects are drawn by different types of lines. Each line has a definite meaning and sense to convey.

➤ **Visible Outlines, Visible Edges:** (Continuous wide lines) the lines drawn to represent the visible outlines/ visible edges / surface boundary lines of objects should be outstanding in appearance.

➤ **Dimension Lines (Continuous Narrow Lines):** Dimension Lines are drawn to mark dimension.

➤ **Extension Lines (Continuous Narrow Lines):** There are extended slightly beyond the respective dimension lines.

➤ **Construction Lines (Continuous Narrow Lines):** These are drawn for

constructing drawings and should not be erased after completion of the drawing.

➤ **Hatching / Section Lines (Continuous Narrow Lines):** These are drawn for the sectioned portion of an object. These are drawn inclined at an angle of 45° to the axis or to the main outline of the section.

➤ **Guide Lines (Continuous Narrow Lines):** These are drawn for lettering and should not be erased after lettering.

➤ **Break Lines (Continuous Narrow Freehand Lines):** Wavy continuous narrow line drawn freehand is used to represent break of an object.

➤ **Break Lines (Continuous Narrow Lines with Zigzags):** Straight continuous narrow line with zigzags is used to represent break of an object.

➤ **Dashed Narrow Lines (Dashed Narrow Lines):** Hidden edges / Hidden outlines of objects are shown by dashed lines of short dashes of equal lengths of about 3 mm, spaced at equal distances of about 1 mm. The points of intersection of these lines with the outlines / another hidden line should be clearly shown.

➤ **Centre Lines (Long-Dashed Dotted Narrow Lines):** These are drawn at the center of the drawings symmetrical about an axis or both the axes. These are extended by a short distance beyond the outline of the drawing.

➤ **Cutting Plane Lines:** Cutting Plane Line is drawn to show the location of a cutting plane. It is long-dashed dotted narrow line, made wide at the ends, bends, and change of direction. The direction of viewing is shown by means of arrows resting on the cutting plane line.

➤ **Border Lines:** Border Lines are continuous wide lines of minimum thickness 0.7 mm.

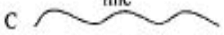
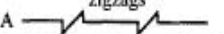
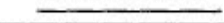
No.	Line description and Representation	Applications
01.1	Continuous narrow line B 	Dimension lines, Extension lines Leader lines, Reference lines Short centre lines Projection lines Hatching Construction lines, Guide lines Outlines of revolved sections Imaginary lines of intersection
01.1	Continuous narrow freehand line C 	Preferably manually represented termination of partial or interrupted views, cuts and sections, if the limit is not a line of symmetry or a center line ⁴ .
01.1	Continuous narrow line with zigzags A 	Preferably mechanically represented termination of partial or interrupted views, cuts and sections, if the limit is not a line of symmetry or a center line ³ .
01.2	Continuous wide line 	Visible edges, visible outlines Main representations in diagrams, maps, flow charts
02.1	Dashed narrow line D 	Hidden edges Hidden outlines
04.1	Long-dashed dotted narrow line E 	Center lines / Axes, Lines of symmetry Cutting planes (Line 04.2 at ends and changes of direction)
04.2	Long-dashed dotted wide line F 	Cutting planes at the ends and changes of direction outlines of visible parts situated in front of cutting plane

Table.1.2 Line description and representation

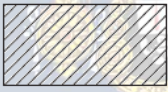
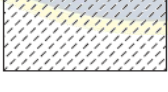
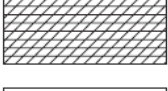
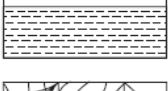
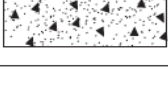
Type	Convention	Material
Metals		Steel, Cast Iron, Copper and its Alloys, Aluminium and its Alloys, etc.
		Lead, Zinc, Tin, White-metal, etc.
Glass		Glass
Packing and Insulating material		Porcelain, Stoneware, Marble, Slate, etc.
		Asbestos, Fibre, Felt, Synthetic resin products, Paper, Cork, Linoleum, Rubber, Leather, Wax, Insulating and Filling materials, etc.
Liquids		Water, Oil, Petrol, Kerosene, etc.
Wood		Wood, Plywood, etc.
Concrete		A mixture of Cement, Sand and Gravel

Table.1.3 Conventional Representation of Materials

1.2 LETTERING

Lettering is defined as writing of titles, sub-titles, dimensions, etc., on a drawing.

1.2.1 IMPORTANCE OF LETTERING:

To undertake production work of an engineering component as per the drawing, the size and other details are indicated on the drawing. This is done in the form of notes and dimensions. Main Features of Lettering are legibility, uniformity, and rapidity of execution. Use of drawing instruments for lettering consumes more time. Lettering should be done freehand with speed. Practice accompanied by continuous efforts would improve the lettering skill and style. Poor lettering mars the appearance of an otherwise good drawing.

1.2.2 SIZE OF LETTERS:

- Size of Letters is measured by the height h of the CAPITAL letters as well as numerals.
- Standard heights for CAPITAL letters and numerals recommended by BIS are given below: 1.8, 2.5, 3.5, 5, 6, 10, 14 and 20 mm.

Note: Size of the letters may be selected based upon the size of drawing.

1.2.3 GUIDE LINES:

To obtain correct and uniform height of letters and numerals, guide lines are drawn, using 2H pencil with light pressure. HB grade conical end pencil is used for lettering. The following are some of the guide lines for lettering.

- Drawing numbers, title block and letters denoting cutting planes, sections are written in 10 mm size.
- Drawing title is written in 7 mm size.
- Hatching, sub-titles, materials, dimensions, notes, etc., are written in 3.5 mm size.
- Space between lines = $\frac{3}{4} h$
- Space between words may be equal to the width of alphabet M or $\frac{3}{5} h$.

1.2.4 PROCEDURE FOR LETTERING:

1. Thin horizontal guide lines are drawn first at a distance ' h ' apart.

2. **Lettering Technique:** Horizontal lines of the letters are drawn from left to right. Vertical, Inclined, and curved lines are drawn from top to bottom.

After lettering has been completed, the guidelines are not erased.

Recommended Size (height h) of Letters / Numerals	
Main Title	5 mm, 7 mm, 10 mm
Sub-Titles	3.5 mm, 5 mm
Dimensions, Notes, etc.	2.5 mm, 3.5 mm, 5 mm

Table.1.4 Sizes for letters

Specifications	Value	Size							
Capital Letter Height	h	2.5	3.5	5	7	10	14	20	
Lowercase Letter Height	$a = (5/7) h$	2.5	3.5	5	7	10	14		
Thickness of Lines	$b = (1/14) h$	0.18	0.25	0.35	0.5	0.7	1	1.4	
Spacing between Characters	$c = (1/7) h$	0.35	0.5	0.7	1	1.4	2	2.8	
Min.Spacing between words	$d = (3/7) h$	1.05	1.5	2.1	3	4.2	6	8.4	
Min. Spacing between Base Lines	$e = (10/7) h$	3.5	5	7	10	14	20	28	

Table.1.5 Specifications for letters

1.2.5 DIMENSIONING:

Drawing of a component, in addition to providing complete shape description, must also furnish information regarding the size description. These are provided through the distances between the Surfaces, location of holes, nature of surface finish, type of material, etc. The expression of these Features on a drawing, using lines, symbols, figures, and notes is called dimensioning.

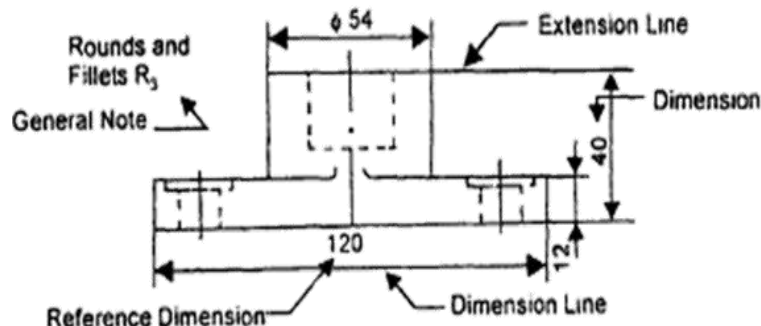


Fig.1.5 Placements on chart

1.2.6 METHODS OF INDICATING DIMENSIONS:

The dimensions are indicated on the drawings according to one of the following two methods.

1.2.6.1 Method - 1 (Aligned method):

Dimensions should be placed parallel to and above their dimension lines and preferably at the middle, and clear of the line. Dimensions may be written so that they can be read from the bottom or from the right side of the drawing. Dimensions on oblique dimension lines should be oriented as shown in Fig.1.6 and except where unavoidable, they shall not be placed in the 30° zone. Angular dimensions are oriented.

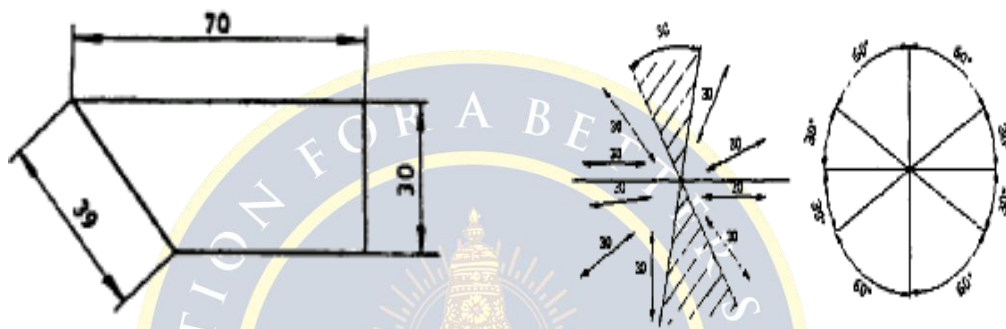


Fig.1.6 Aligned method

1.2.6.2 Method - 2 (Uni-directional):

Dimensions should be indicated so that they can be read from the bottom of the drawing only. Non-horizontal dimension lines are interrupted, preferably in the middle for insertion of the dimension.

Note: Horizontal dimensional lines are not broken to place the dimension in both cases.

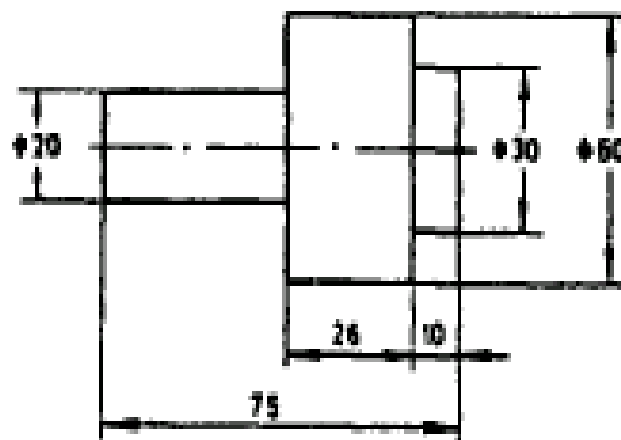


Fig.1.7 Uni-directional method

1.3 GEOMETRICAL CONSTRUCTIONS

1.3.1 Introduction:

Engineering drawing consists of several geometrical constructions. A few methods are illustrated here without mathematical proofs.

1. a) To divide a straight line into a given number of equal parts say 5.

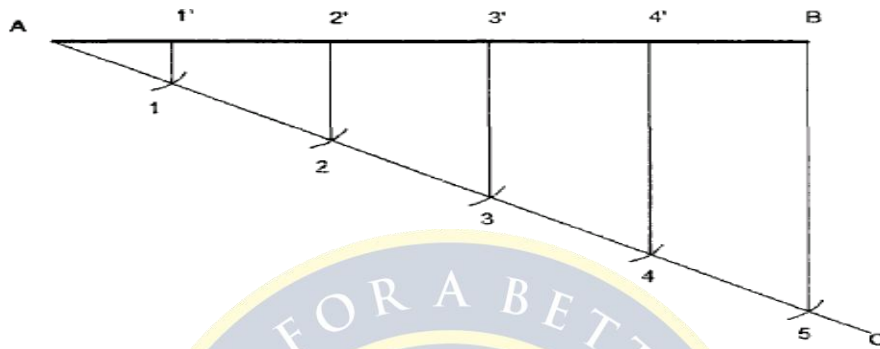


Fig.1.8

Construction:

1. Draw AC at any angle θ to AB.
2. Construct the required number of equal parts of convenient length on AC like 1, 2, 3.
3. Join the last point 5 to B.
4. Through 4, 3, 2, 1 draw lines parallel to 5B to intersect AB at 4', 3', 2' and 1'.

b) To bisect a given angle.

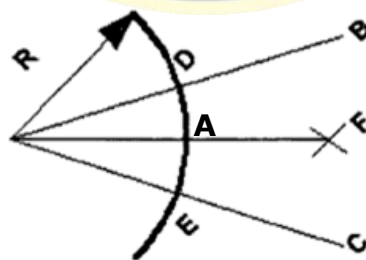


Fig.1.9

Construction:

- a. Draw a line AB and AC making the given angle.
- b. With Centre A and any convenient radius R draw an arc intersecting the sides at D and E.
- c. With Centre's D and E and radius larger than half the chord length DE,

draw arcs intersecting at F.

d. Join AF, $\angle BAF = \angle FAC$.

2. To inscribe a regular polygon of any number of sides in each circle.

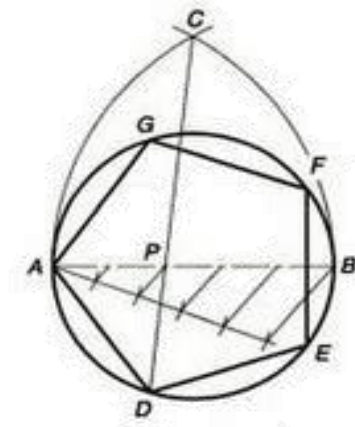


Fig.1.10

Construction:

- Draw the given circle with AB as diameter.
- Divide the diameter AB into N equal parts say 5.
- With AB as radius and A & B as centers, draw arcs intersecting each other at C.
- Join C-P and extend to intersect the circle at D.
- Join A-D which is the length of the side of the required polygon.
- Set the compass to the length AD and starting from D mark off on the circumference of the circles, obtaining the points E, F, etc. The figure obtained by joining the points A, D, E etc., is the required polygon.

3. To inscribe a hexagon in each circle.

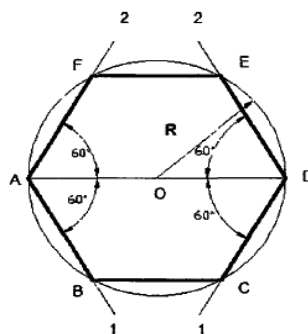


Fig.1.11

Construction:

- With center O and radius R draw the given circle.
- Draw any diameter AD to the circle.

- c. Using 300 - 600 set-square and through the point A draw lines A1, A2 at an angle 60° with AD, intersecting the circle at B and F, respectively.
- d. Using 300-600 set-square and through the point D draw lines D1, D2 at an angle 60° with DA, intersecting the circle at C and E, respectively.
- e. By joining A, B, C, D, E, F and A, S the required hexagon is obtained.

4. To construct a regular polygon (say a pentagon) given the length of the side.

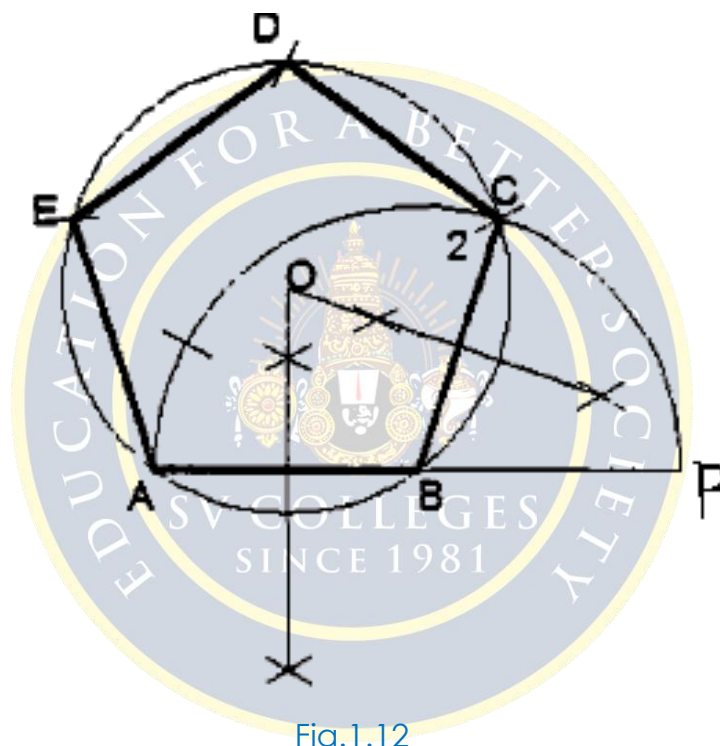


Fig.1.12

Construction:

- a. Draw a line AB equal to the side and extend to P such that $AB = BP$.
- b. Draw a semicircle on AP and divide it into 5 equal parts by trial and error.
- c. Join B to second division. Irrespective of the number of sides of the polygon B is always joined to the second division.
- d. Draw the perpendicular bisectors of AB and B2 to intersect at O.
- e. Draw a circle with O as center and OB as radius.
- f. With AB as radius intersect the circle successively at D and E. Then join CD, DE, and EA.

5. To construct a regular polygon (say a hexagon) given the side AB .

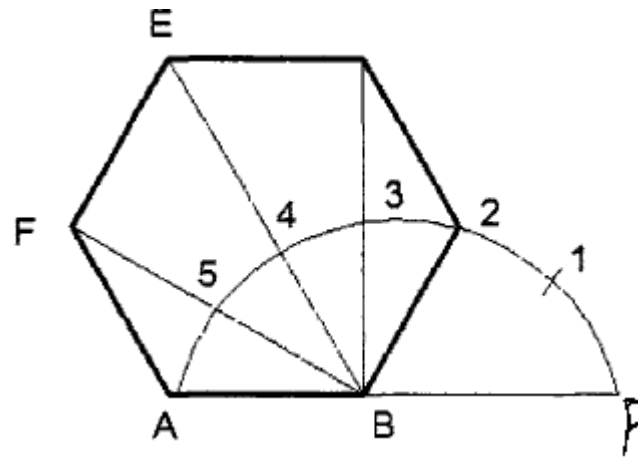


Fig.1.13

Construction:

- Draw a line AB equal to the side and extend to P such that $AB = BP$.
- Draw a semicircle on AP and divide it into 6 equal parts by trial and error.
- Join B to second division.
- Join B-3, B-4, B-5 and produce them.
- With 2 as Centre and radius AB intersect the line B, 3 produced at D. Similarly get the point E and F.
- Join D-E, E-F and F-A to get the required hexagon.

6. To construct a regular figure of given side length and of N sides on a straight line.

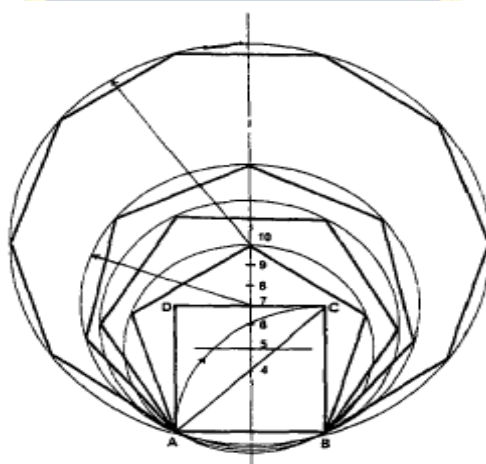


Fig.1.14

Construction:

- Draw the given straight line AB.
- At B erect a perpendicular BC equal in length to AB.

- c. Join AC and where it cuts the perpendicular bisector of AB, number the point 4.
- d. Complete the square ABCD of which AC is the diagonal.
- e. With radius AB and center B describe arc AC as shown.
- f. Where this arc cuts the vertical Centre line numbers the point 6.
- g. This is the Centre of a circle inside which a hexagon of side AB can now be drawn.
- h. Bisect the distance 4-6 on the vertical Centre line.
- i. Mark this bisection 5. This is the Center in which a regular pentagon of side AB can now be drawn.
- j. On the vertical Centre line step off from point 6 a distance equal in length to the distance 5-6. This is the Center of a circle in which a regular heptagon of side AB can now be drawn.
- k. If further distances 5-6 are now stepped off along the vertical center line and are numbered consecutively, each will be the center of a circle in which a regular polygon can be inscribed with side of length AB and with several sides denoted by the number against the center.

1.4 CONIC SECTIONS

Cone is formed when a right angled triangle with an apex and angle θ is rotated about its altitude as the axis. The length or height of the cone is equal to the altitude of the triangle and the radius of the base of the cone is equal to the base of the triangle. The apex angle of the cone is 2θ . When a cone is cut by a plane, the curve formed along the section is known as a conic.

- a) **CIRCLE:** When a cone is cut by a section plane A-A making an angle $\alpha = 90^\circ$ with the axis, the section obtained is a circle.
- b) **ELLIPSE:** When a cone is cut by a section plane B-B at an angle, α more than half of the apex angle i.e., θ and less than 90° , the curve of the section is an ellipse. Its size depends on the angle α and the distance of the section plane from the apex of the cone.
- c) **PARABOLA:** If the angle α is equal to θ i.e., when the section plane C-C is parallel to the slant side of the cone the curve at the section is a parabola. This is not a closed figure like circle or ellipse. The size of the parabola

depends upon the distance of the section plane from the slant side of the cone.

d) **HYPERBOLA**: If the angle α is less than θ (section plane D-D), the curve at the section is hyperbola. The curve of intersection is hyperbola, even if $\alpha = \theta$, provided the section plane is not passing through the apex of the cone. However, if the section plane passes through the apex, the section produced is an isosceles triangle.

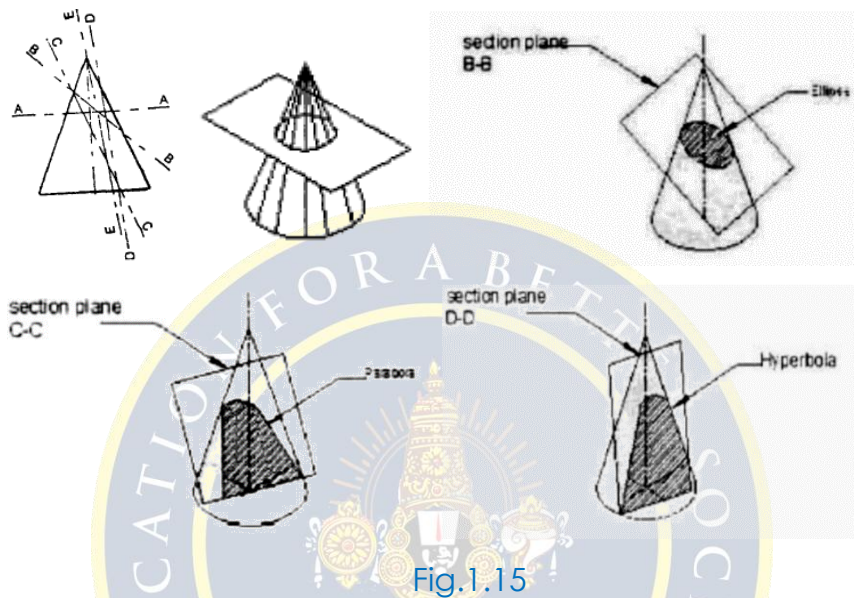


Fig.1.15

- e) Eccentricity:
- a. If $e=1$, it is parabola
 - b. If $e>1$, it is hyperbola
 - c. If $e<1$, it is an ellipse

Where eccentricity e is the ratio of distance of the point from the focus to the distance of the point from the directrix.

e) **PARABOLA**: In physical world, parabola are found in the main cables on simple suspension bridge, as parabolic reflectors in satellite dish antennas, vertical curves in roads, trajectory of a body, automobile head light, parabolic receivers.



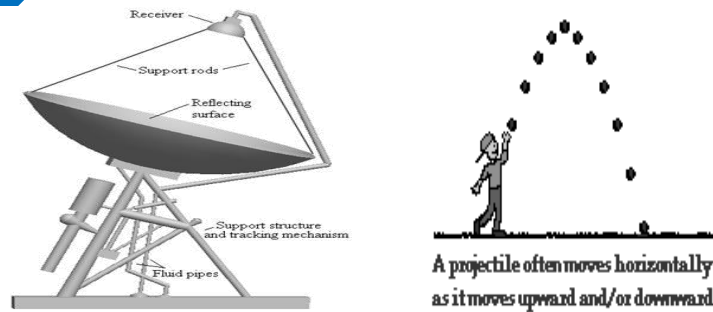


Fig.1.16

1. To draw a parabola with the distance of the focus from the directrix at 50mm (Eccentricity method)

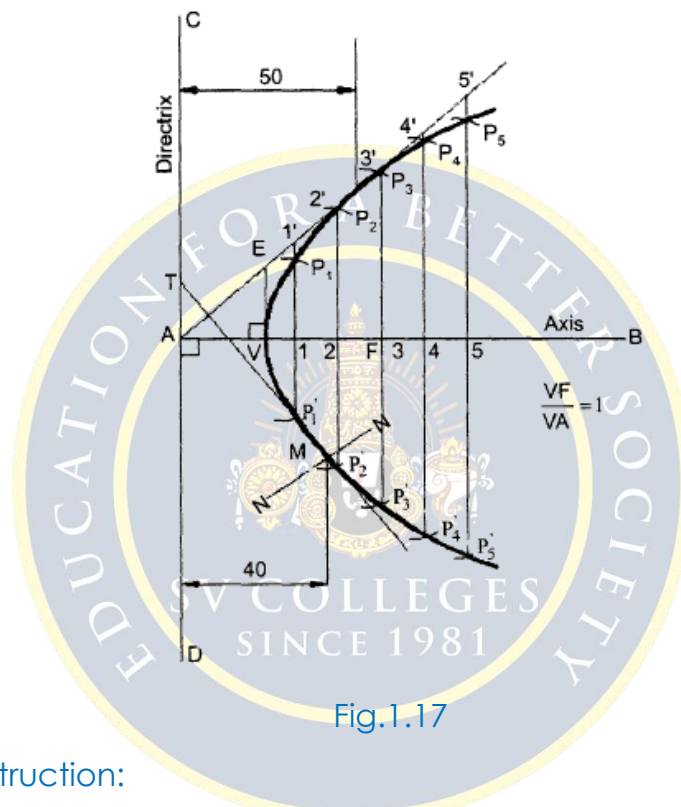


Fig.1.17

Construction:

- Draw the axis AB and the directrix CD at right angles to it:
- Mark the focus F on the axis at 50mm.
- Locate the vertex V on AB such that $AV = VF$
- Draw a line VE perpendicular to AB such that $VE = VF$
- Join A, E and extend. Now, $VE/VA = VF/VA = 1$, the eccentricity.
- Locate number of points 1, 2, 3, etc., to the right of V on the axis, which need not be equidistant.
- Through the points 1, 2, 3, etc., draw lines perpendicular to the axis and to meet the line AE extended at 1', 2', 3' etc.
- With center F and radius 1-1', draw arcs intersecting the line through 1 at P1 and P'1.

- i. Similarly, locate the points P_2, P'_2, P_3, P'_3 etc., on either side of the axis. Join the points by smooth curve, forming the required parabola.

2. To draw a normal and tangent through a point 40mm from the directrix.

To draw a tangent and normal to the parabola. Locate the point M which is at 40 mm from the directrix. Then join M to F and draw a line through F, perpendicular to MF to meet the directrix at T. The line joining T and M and extended is the tangent and a line NN, through M and perpendicular to TM is the normal to the curve.

ELLIPSE:

Ellipses are mostly found as harmonic oscillators, phase visualization, elliptical gears, ellipse wings.

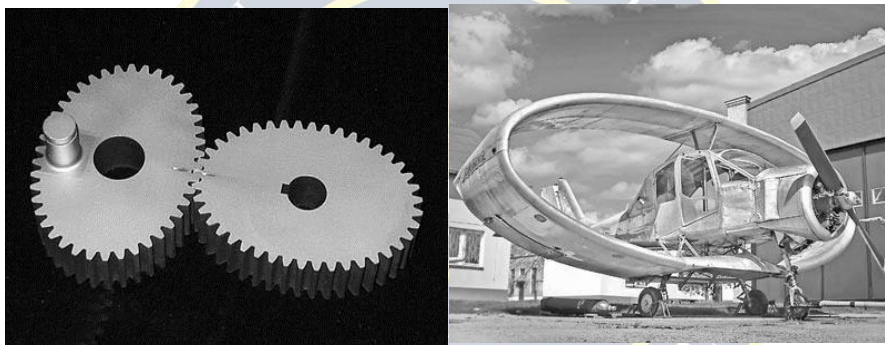


Fig.1.18

3. To draw an ellipse with the distance of the focus from the directrix at 50mm and eccentricity = 2/3 (Eccentricity method)

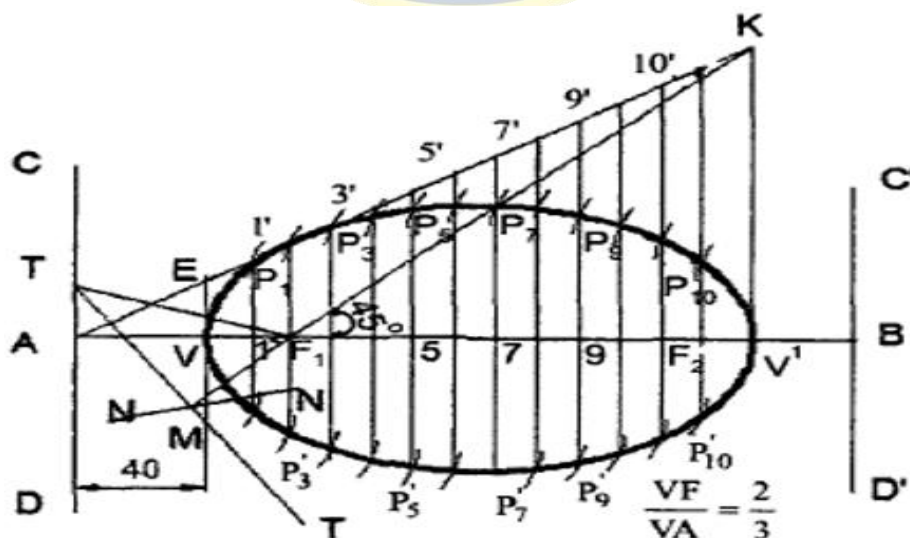


Fig.1.19

Construction:

- Draw any vertical line CD as directrix.
- At any point A in it, draw the axis.
- Mark a focus F on the axis such that $AF = 50\text{mm}$.
- Divide AF in to 5 equal divisions.
- Mark the vertex V on the third division-point from A.
- Thus, eccentricity $e = VF/VA = 2/3$.
- A scale may now be constructed on the axis which will directly give the distances in the required ratio.
- At V, draw a perpendicular $VE = VF$. Draw a line joining A and E.
- Mark any point 1 on the axis and through it draw a perpendicular to meet AE produced at 1'.
- With Centre F and radius equal to 1-1', draw arcs to intersect a perpendicular through 1 at points P1 and P'1.
- Similarly mark points 2, 3 etc. on the axis and obtain points P2 and P'2, P3 and P'3, etc.
- Draw the ellipse through these points, it is a closed curve two foci and two directrices.

1.5 HYPERBOLA

Lampshades, gear transmission, cooling towers of nuclear reactors are some of the applications of Hyperbola.



Fig.1.20

- To draw a hyperbola with the distance of the focus from the directrix at 50mm and $e = 3/2$ (Eccentricity method).

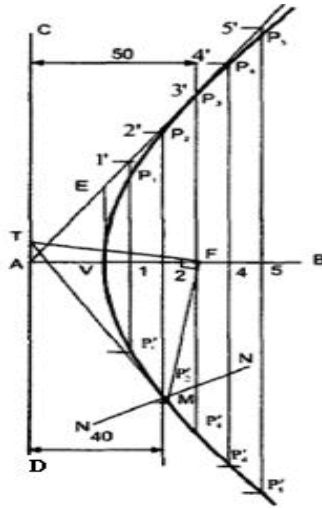


Fig.1.21

Construction:

- Draw the directrix CD and the axis AB.
- Mark the focus F on AB and 65mm from A.
- Divide AF into 5 equal divisions and mark V the vertex, on the second division from A.
- Draw a line VE perpendicular to AB such that $VE = VF$. Join A and E.
- Mark any point 1 on the axis and through it, draw a perpendicular to meet AE produced at 1'.
- With center F and radius equal to $F1'$, draw arcs intersecting the perpendicular through 1 at P1 and P'1.
- Similarly mark several points 2, 3 etc., and obtain points P2 and P'2, etc.

1.6 SPECIAL CURVES

1.6.1 CYCLOIDAL CURVES:

Cycloidal curves are generated by a fixed point in the circumference of a circle when it rolls without slipping along a fixed straight line or circular path. The rolling circle is called the generating circle, the fixed straight line, the directing line and the fixed circle, the directing circle.

In physical world, cycloidal curves are used as cycloidal gears, epicyclic train dynamometer, epicyclic gear train, hypocycloid engine.

1.6.2 CYCLOID:

A cycloid is a curve generated by a fixed point on the circumference of a circle when it rolls without slipping along a straight line.

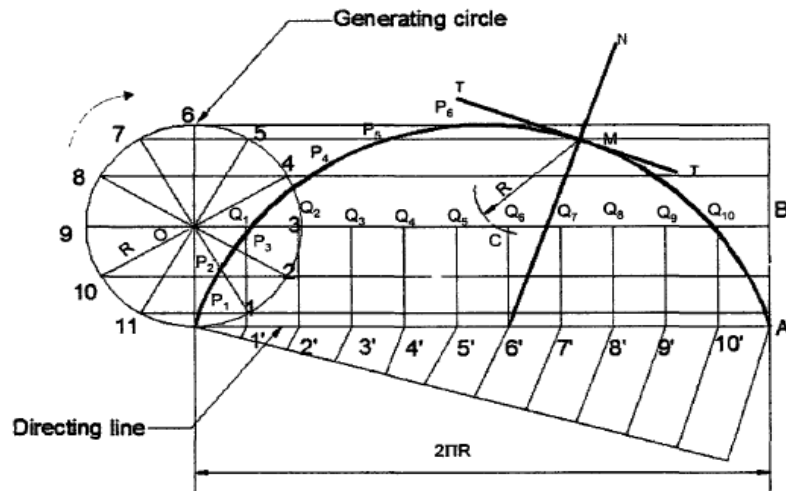
1. To draw a cycloid, given the radius R of the generating circle.

Fig.1.22

Construction:

- With center O and radius R , draw the given generating circle.
- Assuming point P to be the initial position of the generating point, draw a line PA , tangential and equal to the circumference of the circle.
- Divide the line PA and the circle into the same number of equal parts and number the points.
- Draw the line OB , parallel and equal to PA . OB is the locus of the center of the generating Circle.
- Erect perpendiculars at $1', 2', 3'$, etc., meeting OB at Q_1, Q_2, Q_3 etc.
- Through the points $1, 2, 3$ etc., draw lines parallel to PA .
- With center O , and radius R , draw an arc intersecting the line through 1 at P_1 , P_1 is the position of the generating point, when the center of the generating circle moves to Q_1 .
- Similarly locate the points P_2, P_3 etc.
- A smooth curve passing through the points P, P_1, P_2, P_3 etc., is the required cycloid.

To draw a normal and tangent to a cycloid

- Mark a point M on the cycloid at a given distance from the directing line.
- With M as a center and the radius R , cut the center line at point C .

3. Through point C, draw a line perpendicular to PA, which meets PA at Point NI.
4. Join NIM and extend it to N. The line NNI is the required normal.
5. Through Point M, draw a line TTI Perpendicular to NNI. The line TTI is the required tangent.

1.6.3 EPICYCLOID:

An epi-cycloid is a curve traced by a point on the circumference of a generating circle when it rolls without slipping on another circle (directing circle) outside it.

2. To draw an epi-cycloid, given the radius 'r' of the generating circle and the radius 'R' of the directing circle.

Construction:

- a. With center O' and radius R, draw a part of the directing circle.
- b. Draw the generating circle, by locating the center O of it, on any radial line O' P extended such that $OP = r$.
- c. Assuming P to be the generating point, locate the point, A on the directing circle such that the arc length PA is equal to the circumference of the generating circle. The angle subtended by the arc PA at O' is given by $\theta = \text{angle } \angle A = r * 3600$
- d. With center O' and radius O'O, draw an arc intersecting the line O'A produced at B. The arc OB is the locus of the center of the generating circle.
- e. Divide the arc PA and the generating circle into the same number of equal parts and number the points.
- f. Join O'-1', O'-2', etc., and extend to meet the arc OB at O1, O2 etc.
- g. Through the points 1, 2, 3 etc., draw circular arcs with O' as center.
- h. With center O1 and radius r, draw an arc intersecting the arc through 1 at P1.
- i. Similarly, locate the points P2, P3 etc.
- j. A smooth curve through the points P1, P2, P3 etc., is the required epi-cycloid.

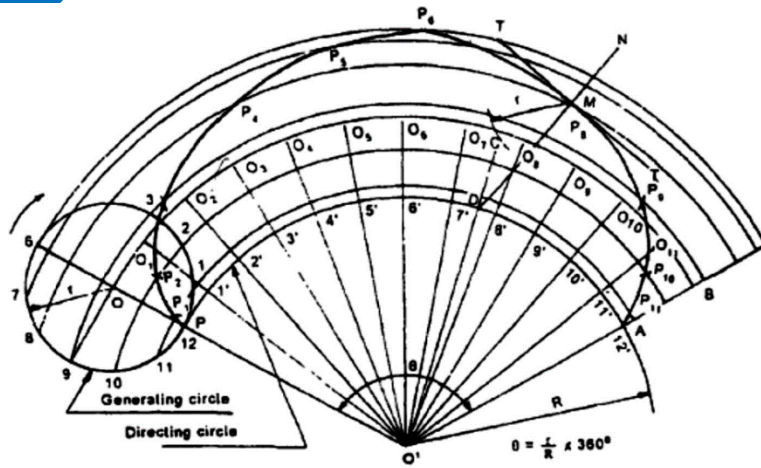


Fig.1.23

To draw a normal and tangent to an Epicycloid.

- Mark a point M on the epicycloid at a given distance from the Point O'.
- With M as the center and radius r, cut the center arc OB at point C.
- Join O'C to meet the arc PA at point D.
- Join DM and Produce it to N. The line DN is the required normal.
- Through Point M, Draw a line TTl Perpendicular to DN. The line TTl is the required tangent.

1.6.4 HYPOCYCLOID:

If the generating circle rolls inside the directing circle, the curve traced by the point is called hypo- cycloid.

- Draw a hypocycloid of a circle of 40 mm diameter which rolls inside another circle of 200 mm diameter for one revolution. Draw a tangent and normal at any point on it.

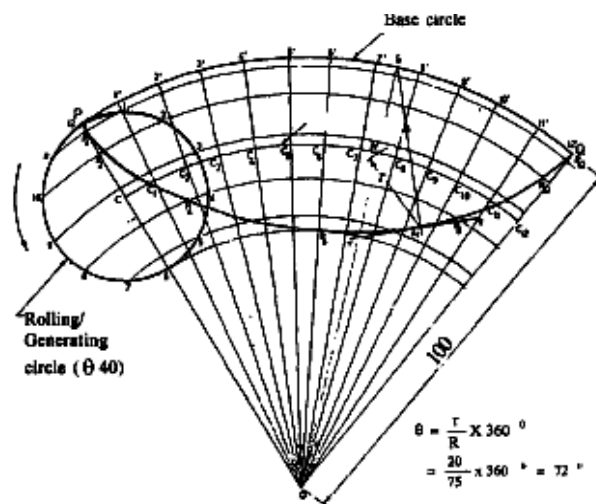


Fig.1.24

Construction:

- a. Taking any point O as center and radius (R) 100 mm draw an arc PQ which subtends an angle $\theta = 72^\circ$ at O. $\theta = \text{angle} = \frac{r}{R} \times 3600$
- b. Let P be the generating point. On OP mark PC = r = 20 mm, the radius of the rolling circle.
- c. With C as center and radius r (20 mm) draw the rolling circle. Divide the rolling circle into 12 equal parts as 1,2,3 etc., in clock wise direction since the rolling circle is assumed to roll counter clock wise.
- d. With O as center, draw concentric arcs passing through 1, 2, 3 etc.
- e. With O as center and OC as radius draw an arc to represent the locus of center.
- f. Divide the arc PQ into same number of equal parts (12) as 1', 2', 3' etc.
- g. Join O'1, O'2 etc., which intersect the locus of center at C1C2C3 etc.
- h. Taking center C1 and radius r, draw an arc cutting the arc through 1 at P1. Similarly obtain the other points and draw a smooth curve through them.
- i. To draw a tangent and normal at a given point M:
- j. With M as center and radius r = CP cut the locus of center at the point N.
- k. Join ON and extend it to intersect the base circle at S.
- l. Join MS, the normal.
- m. At M, draw a line perpendicular to MS to get the required tangent.

Example : Draw an epicycloid of rolling circle of diameter 40 mm which rolls outside another circle (base circle) of 150 mm diameter for one revolution. Draw a tangent and normal at any point on the curve.

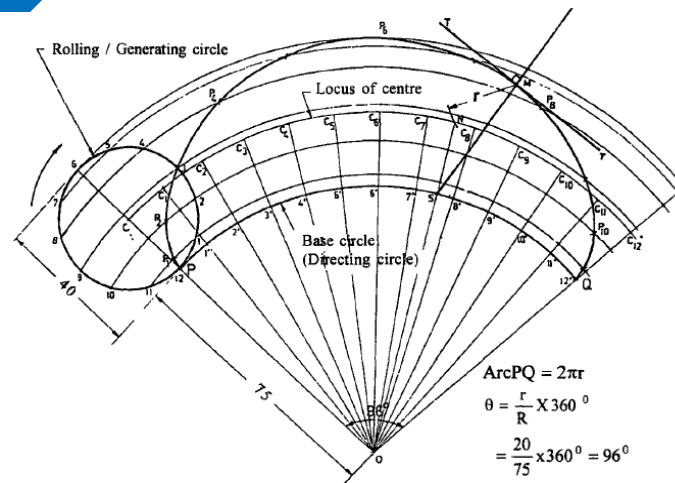


Fig.1.25

$$\theta = \frac{r}{R} \times 360^\circ$$

1.6.5 INVOLUTES:

1. Draw the involute of an equilateral triangle of side 20 and draw a normal and a tangent at a distance 60 from the center of the triangle.

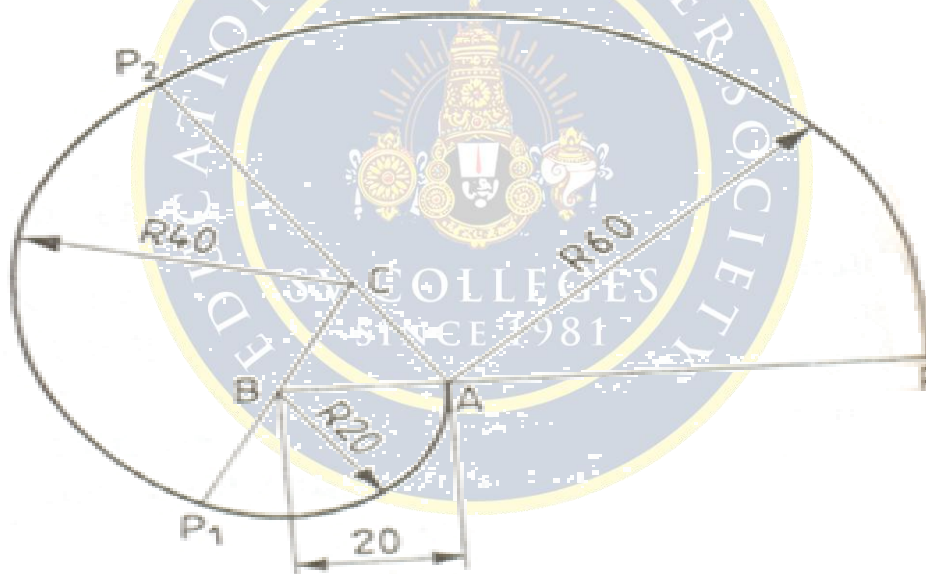


Fig.1.26

- a. Draw the given triangle ABC of side 20.
- b. Assuming A as the starting point; with B as center and radius BA(=20), draw an arc intersecting the line CB extended at P1.
- c. With center C and radius CP1 (=2 x 20), draw an arc intersecting the line AC extended at P2.
- d. With center A and radius AP2 (= 3 x 20), draw an arc intersecting the line BA produced at P3.
- e. The curve through A, P1, P2, P3 is the required involute.

2. Draw the involute of a regular hexagon of side 20. Draw a tangent and a normal to the curve at a distance 100 from the center of the hexagon.

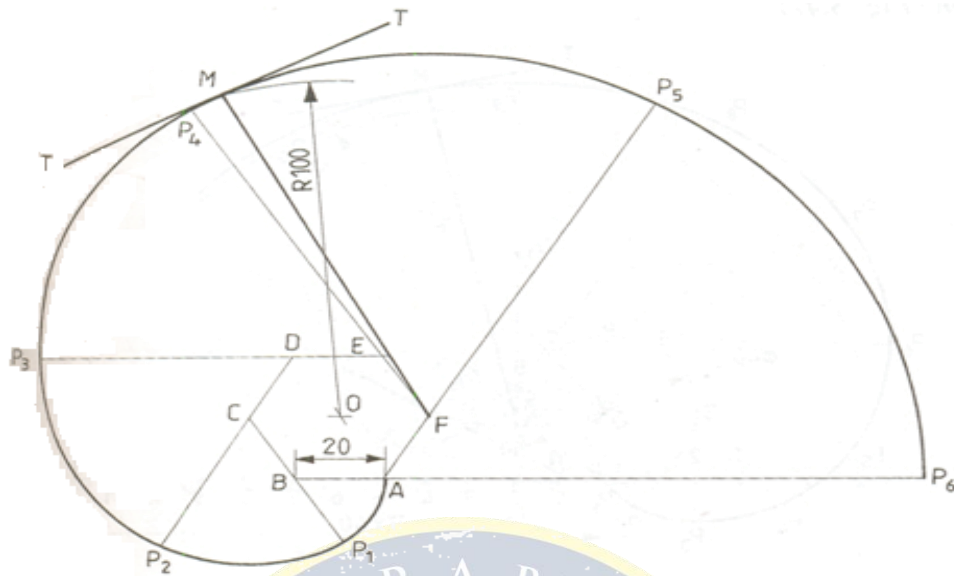


Fig.1.27

- Draw the hexagon ABCDEF of side 20 and locate its center O. Assuming that the thread is unwound from A in the clock-wise direction, the starting point for the involute is A.
- With center B and radius BA (=20), draw an arc intersecting the line CB extended at P1.
- With center C and radius CP1 (=2 x 20), draw an arc intersecting the line DC extended at P2.
- In a similar way, obtain the other points P3, P4, etc.

A smooth curve through the above points is the required involute.

3. Draw the involute of a circle of 40 diameter. Also, draw a tangent and a normal to the curve at a point 95 from the center of the circle.

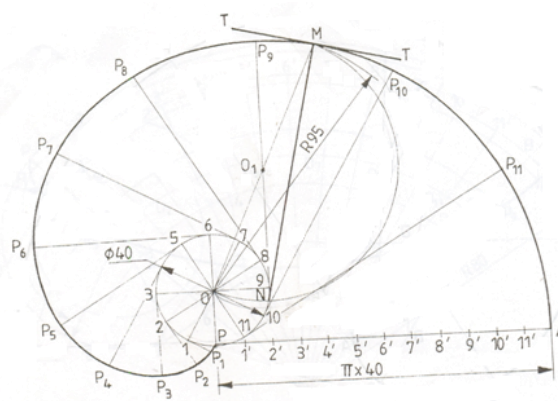


Fig.1.28

- With center O and diameter 40, draw the given circle.
- Assuming P as the starting point, draw a line PA, tangent to the circle and equal to the circumference of the circle.
- Divide the circle and the line PA into the same number of equal parts and number as shown.
- Draw a tangent to the circle at the point 1 and locate on it P1 such that $1P1 = P1'$.
- Draw a tangent to the circle at point 2 and locate on it P2 such that $2P2 = P2'$.
- Locate other points P3, P4, etc., in a similar way.
- A smooth curve through these points is the required involute.
- From the construction, it is obvious that a tangent to the circle is normal to the involute to draw the tangent and normal.
- Locate the point M on the curve, which is at 95 from the center of the circle
- Join M, O and locate its mid-point O1.
- With center O1 and radius O1-M, draw a semi-circle intersecting the given circle at N.

Join N, M forming the normal to the curve and a line T-T, perpendicular to NM at M is the tangent to the curve.

8. Assignments

S.No	Question	BL	CO
1	Construct an ellipse, with distance of the focus from the directrix as 50mm and eccentricity as $2/3$. Also draw normal and tangent to the curve at a point 40mm from the directrix		1
2	Construct a rectangular hyperbola, when a point P on it is at distances of 18 and 34 from two asymptotes. Draw Normal and Tangent at a distance of 25mm from any asymptote.		1
3	Construct a cycloid, given the diameter of the generating circle as 40mm. Draw tangent to the curve at a point on it, 35mm from the directing line.		1
4	Draw an epi – cycloid of a circle of 40mm diameter, which rolls on another circle of 120mm diameter for one revolution in clock – wise. Draw a tangent and a normal to it at a point 90mm from the center of the directing circle.		2
5	A Hexagonal Pyramid side of base 30mm and axis 50mm long rests with one of the corners of its based on H.P. Its axis is inclined at 35° to H.P. and 45° to V.P. Draw its projections.		1

9. Part B- Questions

S.No	Question	BL	CO
1	Construct an ellipse, with distance of the focus from the directrix as 50mm and eccentricity as $\frac{2}{3}$. Also draw normal and tangent to the curve at a point 40mm from the directrix		1
2	Construct a rectangular hyperbola, when a point P on it is at distances of 18 and 34 from two asymptotes. Draw Normal and Tangent at a distance of 25mm from any asymptote.		1
3	Construct a cycloid, given the diameter of the generating circle as 40mm. Draw tangent to the curve at a point on it, 35mm from the directing line.		1
4	Draw an epi – cycloid of a circle of 40mm diameter, which rolls on another circle of 120mm diameter for one revolution in clock – wise. Draw a tangent and a normal to it at a point 90mm from the center of the directing circle.		2
5	A Hexagonal Pyramid side of base 30mm and axis 50mm long rests with one of the corners of its based on H.P. Its axis is inclined at 35° to H.P. and 45° to V.P. Draw its projections.		1

10. Supportive Online Certification Courses

1. Engineering drawing and computer graphics By Prof. Rajaram Lakkaraju, conducted by IIT Kharagpur – 12 weeks

11. Prescribed Textbooks & Reference Books

Textbook

1. K.L. Narayana & P. Kanniah, Engineering Drawing, 3/e, SciTech Publishers, Chennai, 2012.
2. N.D. Bhatt, Engineering Drawing, 53/e, Charotar Publishers, 2016.

References:

1. Venugopal, Engineering Drawing and Graphics, 3/e, New Age Publishers, 2000.
2. K.C. John, Engineering Graphics, 2/e, PHI, 2013.